stabilizer bar is used to make the body of the vehicle will remain flat throughout a turn so the weight stays evenly distributed on all four tires. It is like a U shaped bar which used in front independent suspension system.

7.1.1.4 Torsion Bar

The simple layout of torsion bar is shown in the figure 7.1.1.4. The torsion bar is made up of hardened steel alloys. It works based on the torque principle. It is used in the independent suspension system. The one end of the torsion bar is fixed with frame and another end is fixed with wheel arm. Wheel arm is supported by bearing. Another end of wheel arm is attached with the wheel hub spindle.

While travelling in bumpy road, the vibration absorption takes place by twisting. To fix the torsion bar, a minimum space is required.

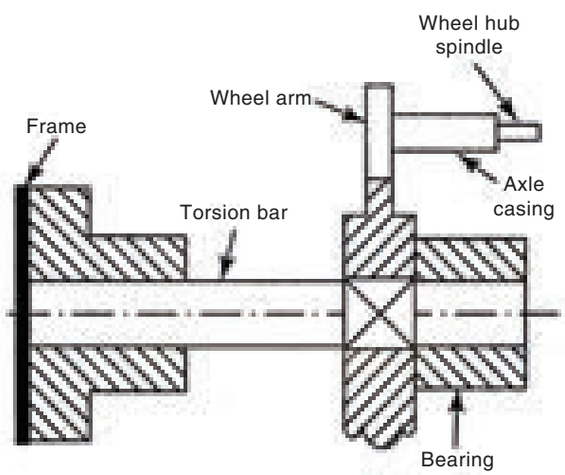


Figure 7.1.1.4 (a) Torsion bar

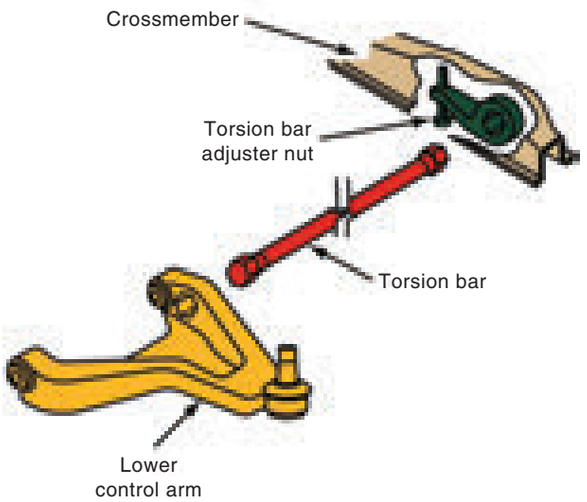


Figure 7.1.1.4 (b) Torsion bar

7.1.2 Rubber Spring

The layout of rubber spring is shown in figure 7.1.2. Rubber springs are used in suspension systems. It absorbs more vibrations than metal springs. There are different types of rubber springs are used.

1. Compression spring
2. Compression shear spring
3. Steel reinforced spring
4. Progressive spring

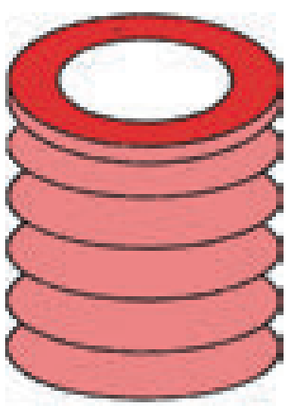


Figure 7.1.2 Rubber Spring

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